

① JDK → JAVA Development Kit

JRE → JAVA runtime environment

IDE → integrated development environment

popular IDE's:

eclipse (very popular)

net-beans

intelliJ idea

Architecture:-

S.O.P.T.H ("hello") → JRE → JAVAC (compiler) → JAR

JRE → JVM

JVM → Java virtual machine

output ← .class ← binary

- JAVA is OOP i.e. every thing is class
- Java should have naming, method to create
- naming is method fashion
- every line ends with ; (semicolon)

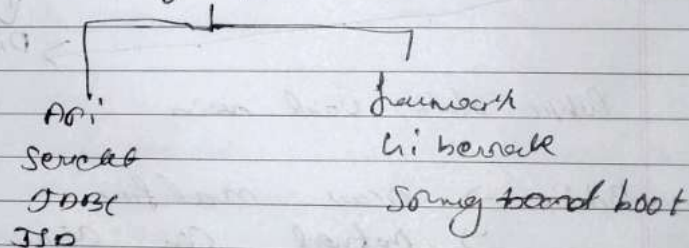
## JAVA full stack.

②

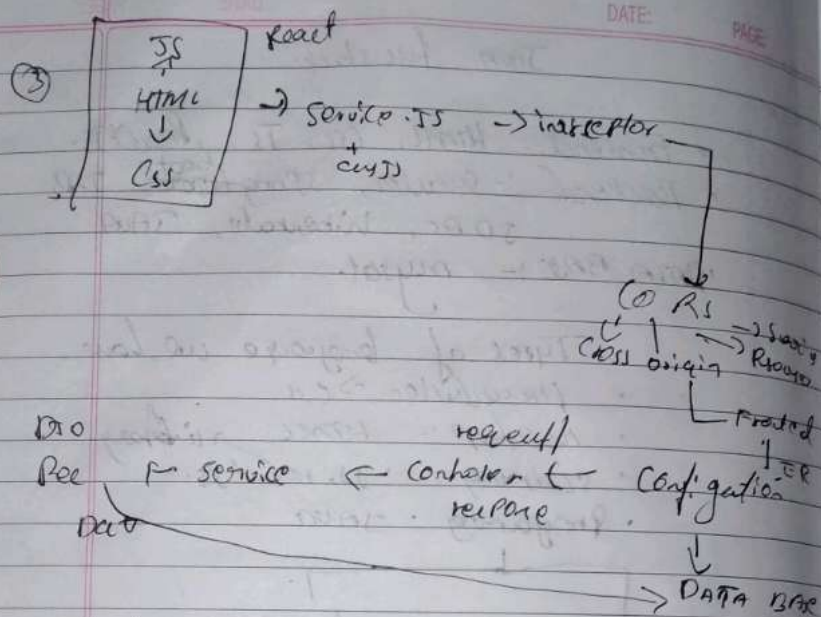
- Frontend:- HTML, CSS, JS, ReactJS.
- Backend :- servlet, spring <sup>boot</sup> ~~boot~~, JSP, JDBC, Hibernate, JPA.
- DATA BASE :- mysql.

### Type of language we have.

- presentation -> CSS
- Markup :- HTML
- scripting :- JS, reactJS.
- Programming :- JAVA



- Query language :- my sql



Public static void main

Public → access modifier

→ method can access everywhere

Static → the method belongs to the class, not an object

Don't can call main() method

So it must be public

without creating an object of the class

void → is return type

→ method does not return anything



DATE: PAGE:  
(4) main var in method name  
→ Switches for a method named  
main to start the program

Stack concept :-

- main on every stage
- top stored toward 100  
address.

• is the main method in Java.  
it is the entry point of a Java  
application.

Primitive

Non Primitive

• Built in data type

• Size depend on object

• Stores actual value

• Slightly slower because  
they involve objects

• Fixed size

• cannot call method

• can call methods  
have properties

Ex:- int, char, float

• Default value is null

Ex:- String, array,

Class.

⑦ a class can be declared with or without

Python

They want in even number  
so it must be greater than  
2 and

```
w = int(input())  
if w > 2 and w % 2 == 0:  
    print("yes")  
else:  
    print("no")
```

```
public class main {  
    Scanner sc = new Scanner(System.in);  
    System.out.println("Day of week");  
    System.out.println("Enter a number");  
    if (dayNumber == 1) {  
        S.o.p("Monday")  
    }  
    else if (dayNumber == 2) {  
        S.o.p("Tuesday")  
    }  
    else {  
        S.o.p("Invalid day")  
    }  
}
```

## ⑥ OOPS

- class : blueprint or object
- object
  - inheritance
  - polymorphism
  - encapsulation
  - abstraction

What is constructor.

and object

• class

class is a blueprint or template used to create object

Object:-

An object is an instance of a class

Constructor:-

a constructor is a special method that is automatically called when an object is created

Type:-

- Default constructor
- Parameterised constructor



⑦ ways to create an object in Java.

- using the (new) keyword.
- using reflection
- using clone()
- using factory method.
- using Object Deserialization.

What is the Object class

The Object class is the root (Super) class of all classes in Java. Every class in Java directly or indirectly inherits from the Object class.

Methods

• toString()

• equals

• hashCode

• getClass

• clone()

• wait()

• notify()

• notifyAll()

Inheritance.

⑧ inheritance is an oop concept in which one class the properties and method of another class sharing the control & behavioral.

Types

1. single inheritance. → inherit from one class
2. multilevel → parameter class, which is its own class
3. hierarchical → inherit from same class
4. multiple → more than one
5. hybrid → two or more

Right click → source → generate data  
set

select all → generate

Encapsulation

it is one of the main in oop in java.  
process of wrapping data and method into  
a single unit.